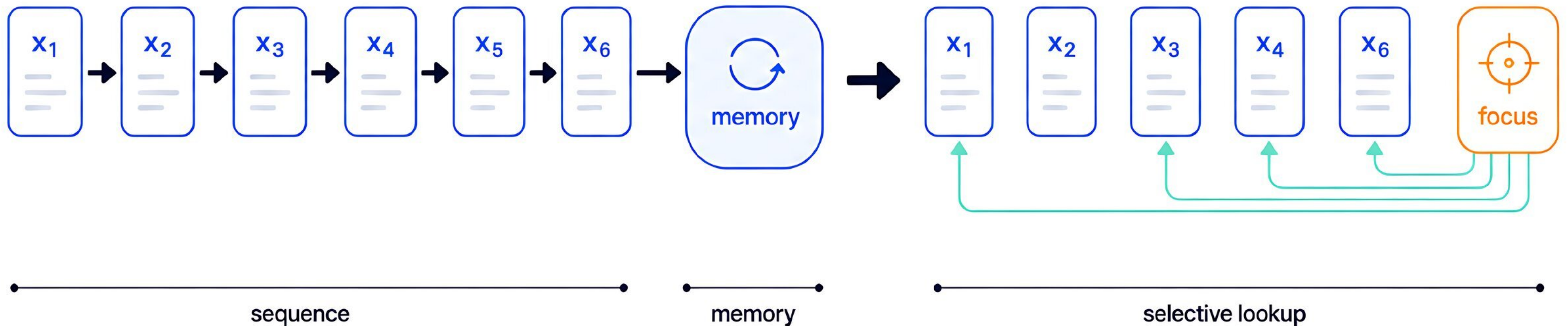


From RNNs to Attention

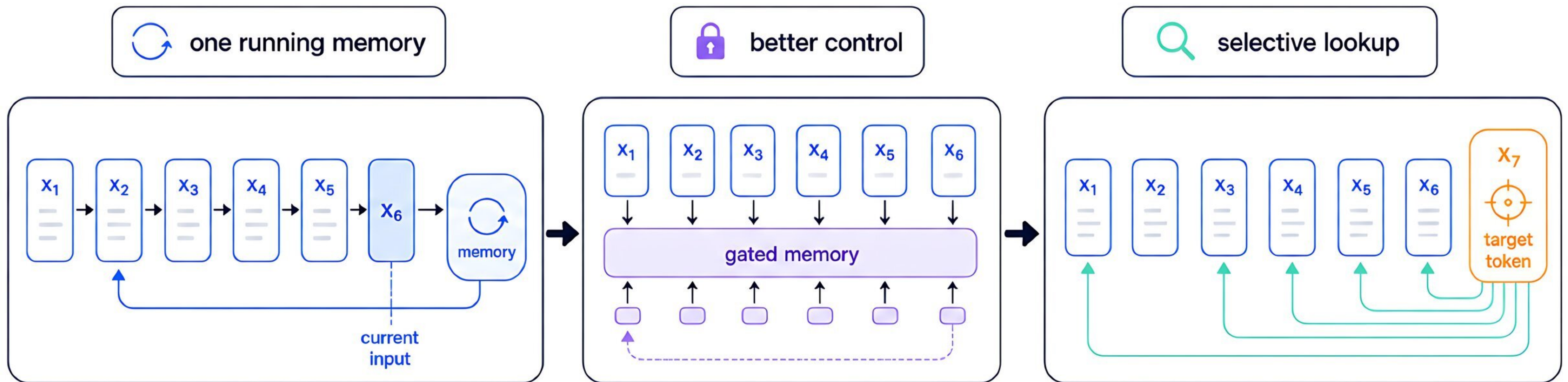
How models moved from carrying memory to looking up context



Parsa Salamatipour

The journey of this lesson

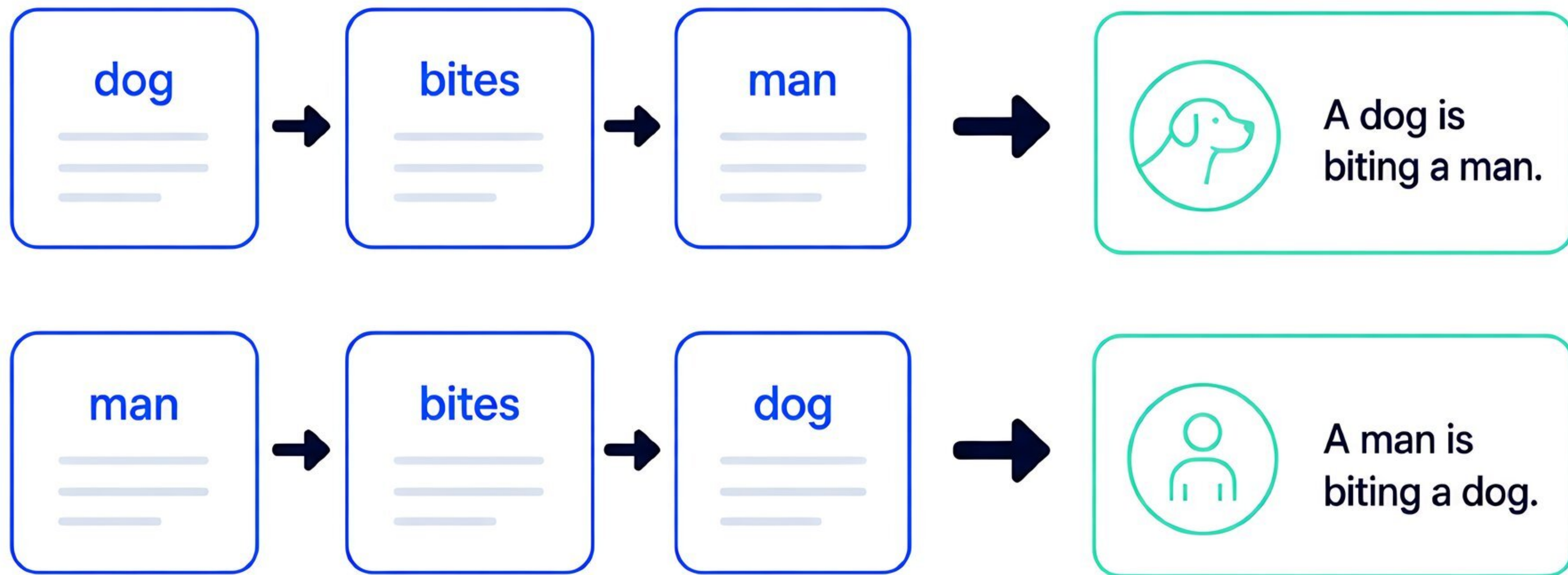
Carry context forward → look useful context up



Today: build the intuition for self-attention.

Why sequences are different

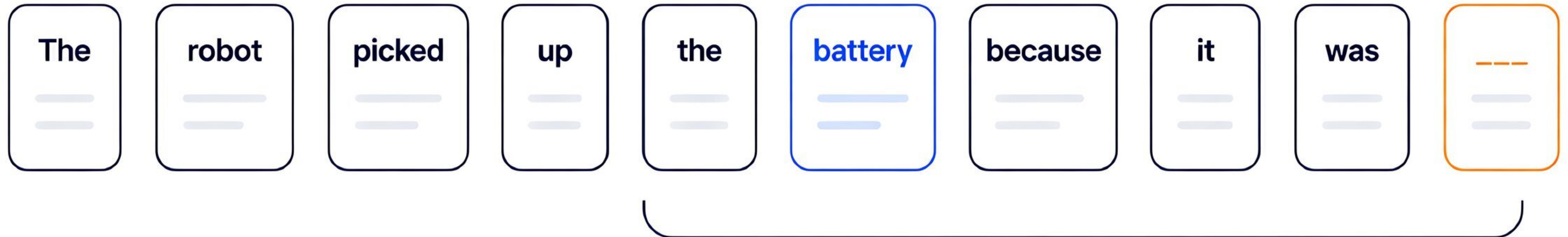
Order changes meaning.



A clue can be far away

Predict the missing word.

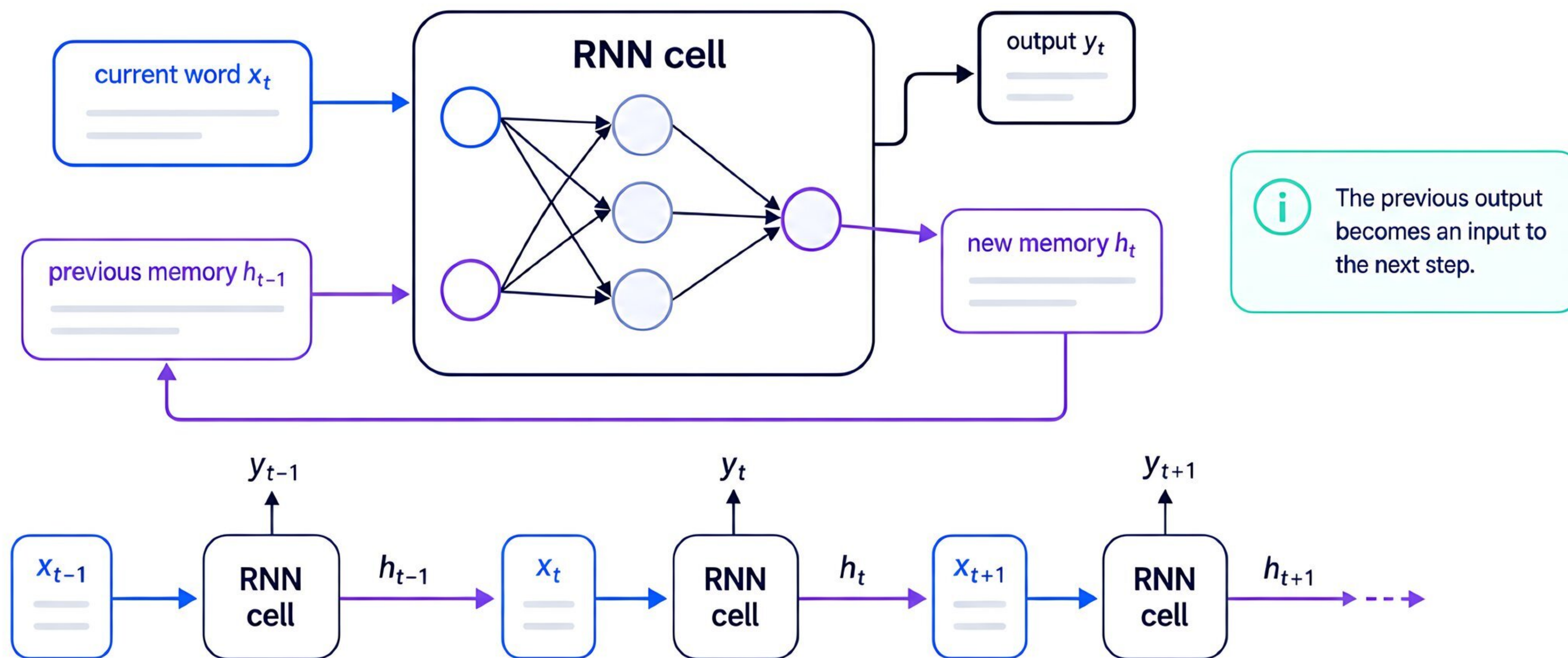
The robot picked up the battery because it was —



What belongs here?

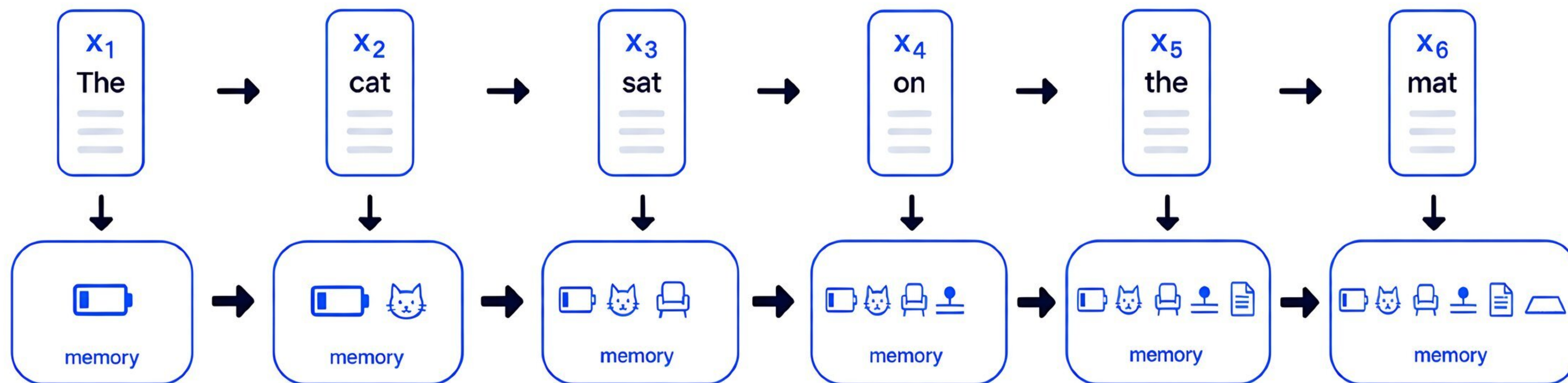
RNN: carry a memory forward

Current input + previous memory $\rightarrow$ new memory



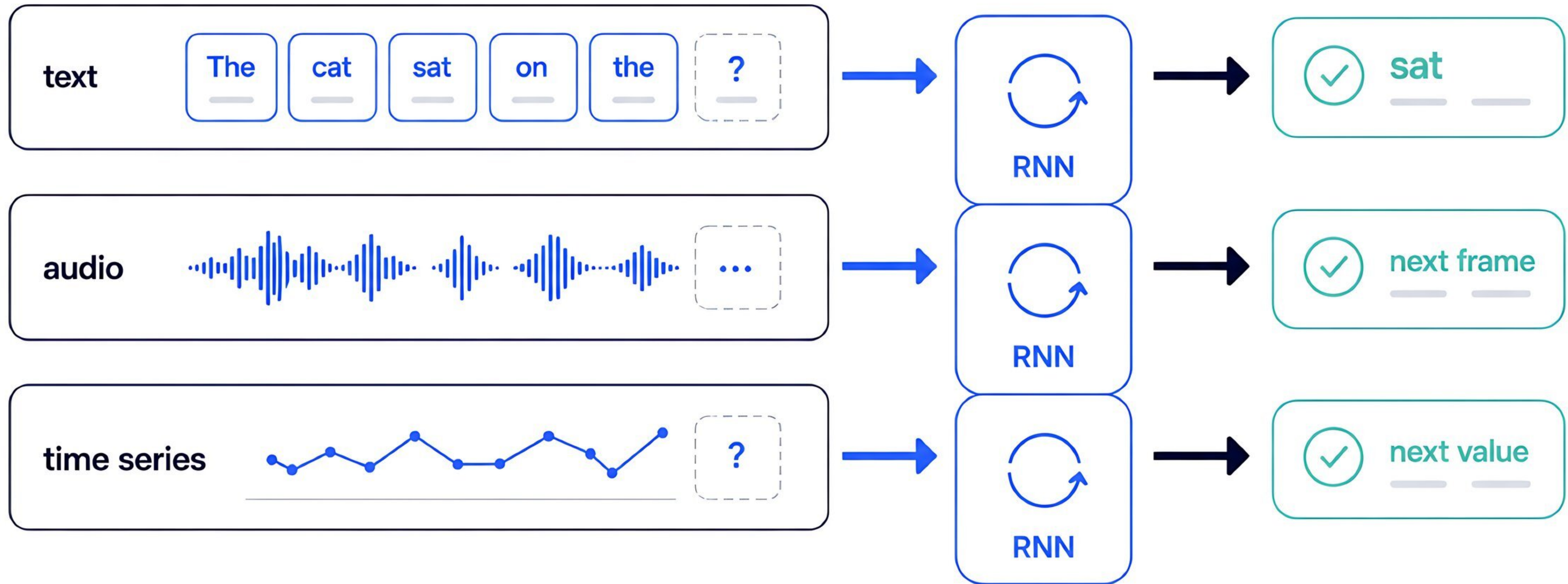
Reading one step at a time

The memory is rewritten at every word.



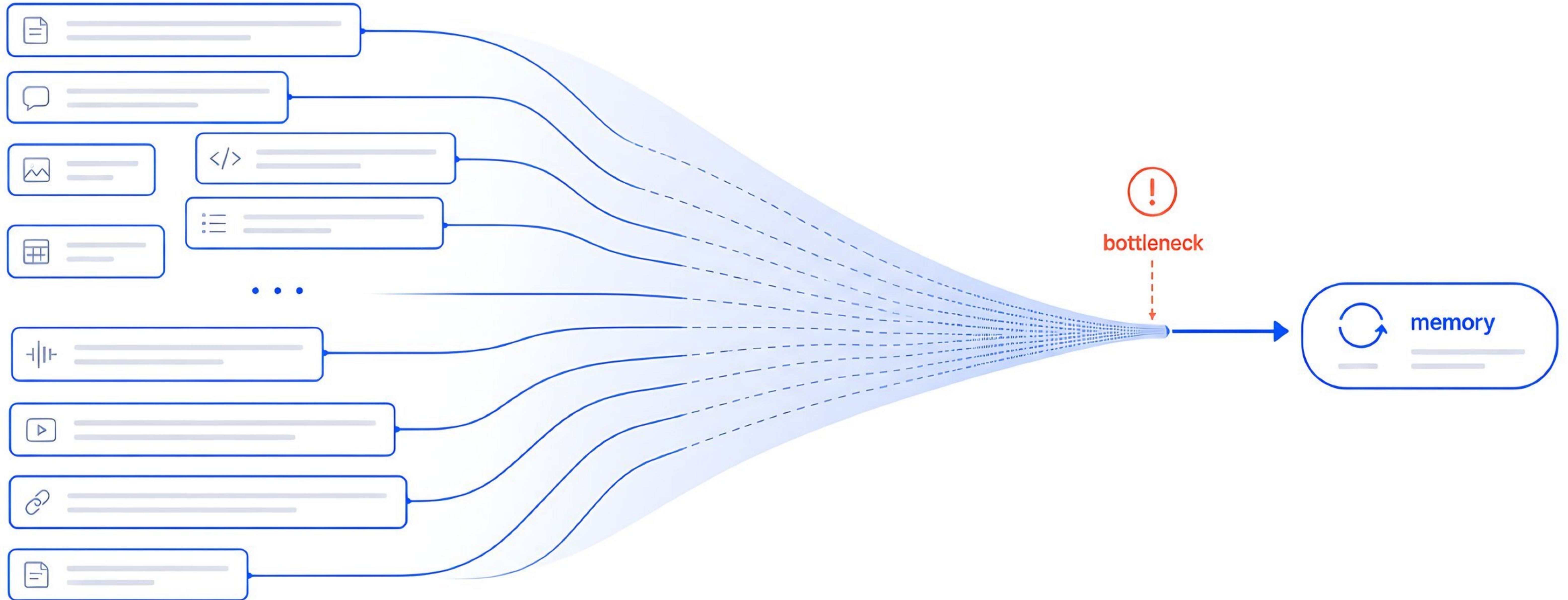
Where RNNs help

Short sequences with local patterns



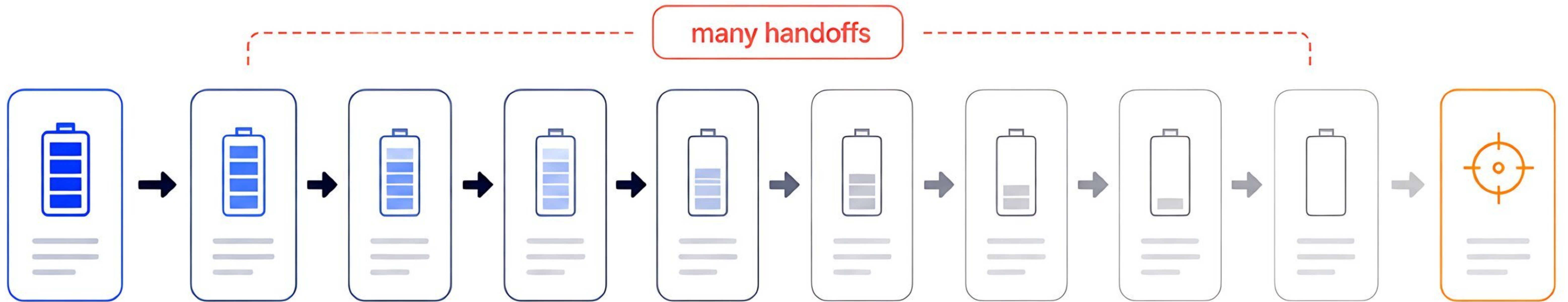
One memory has to carry everything

A long story is squeezed into one running summary.



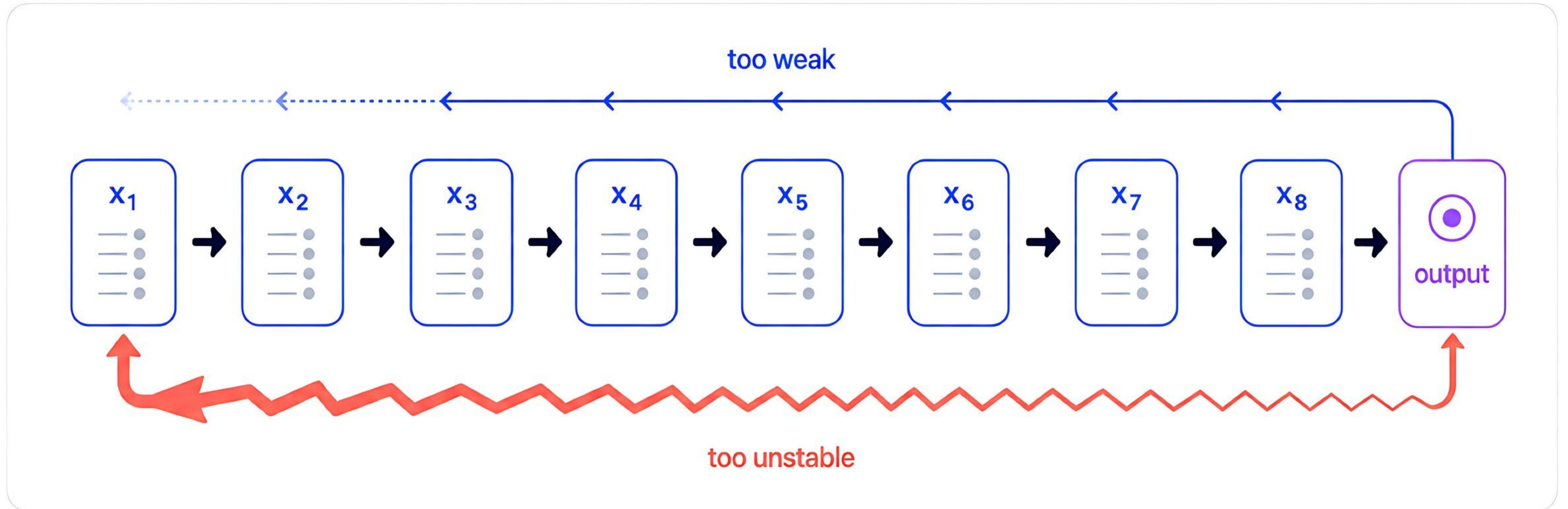
The long-distance problem

Important clues must survive every step.



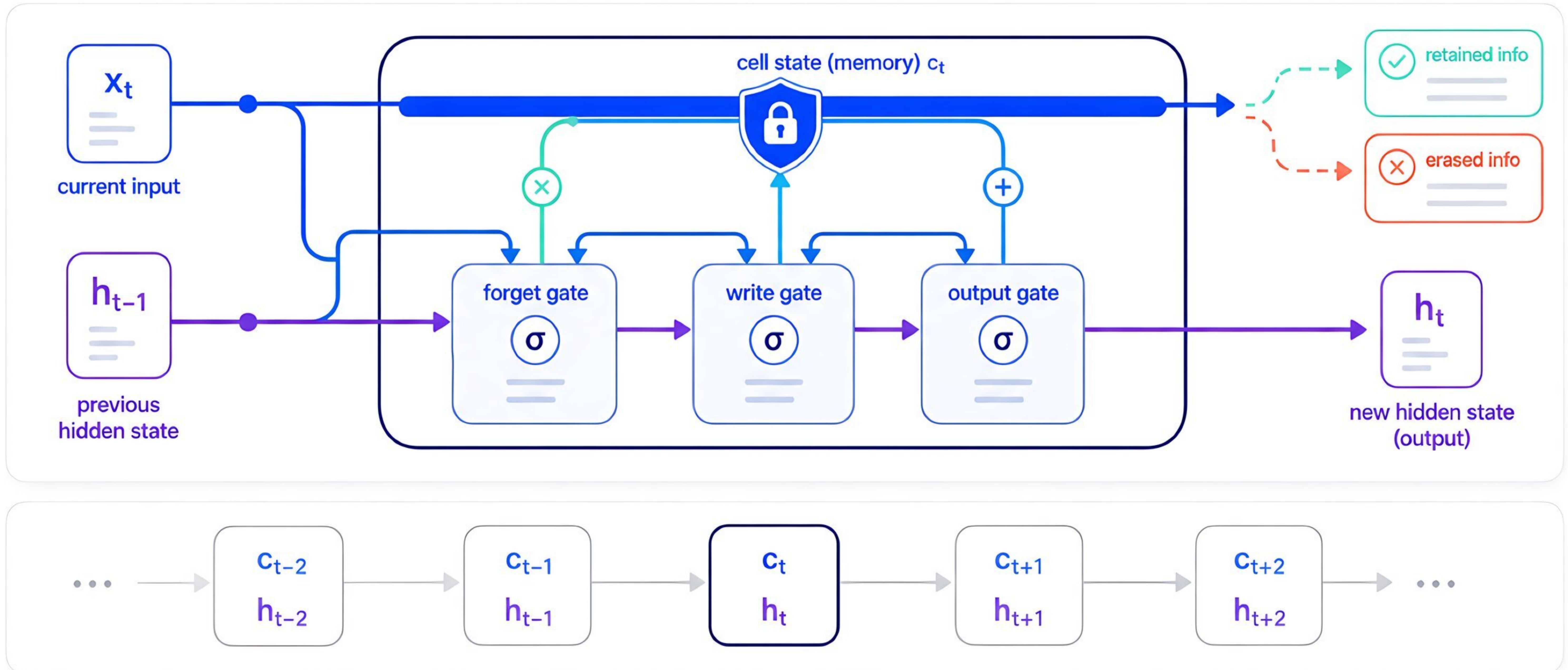
Learning through a long chain

Feedback can fade or become unstable.



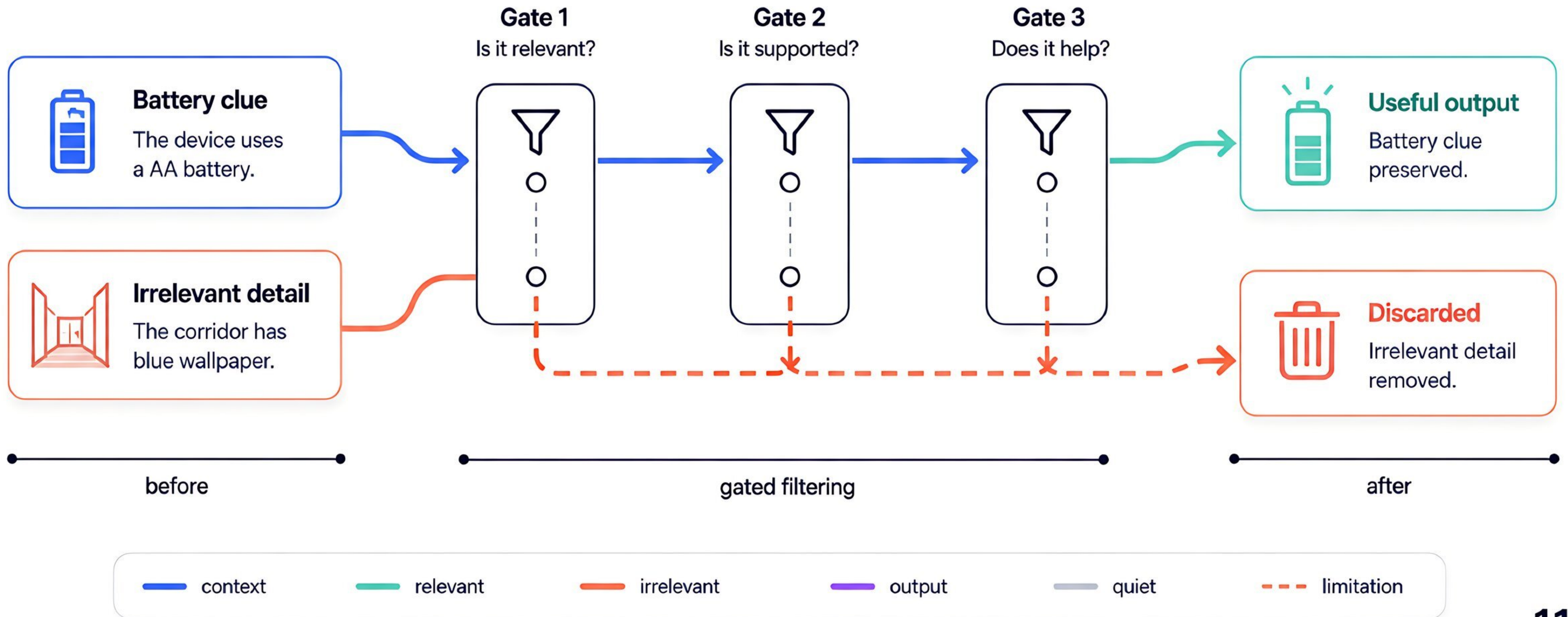
LSTM: memory with controls

Keep, erase, and write information deliberately.



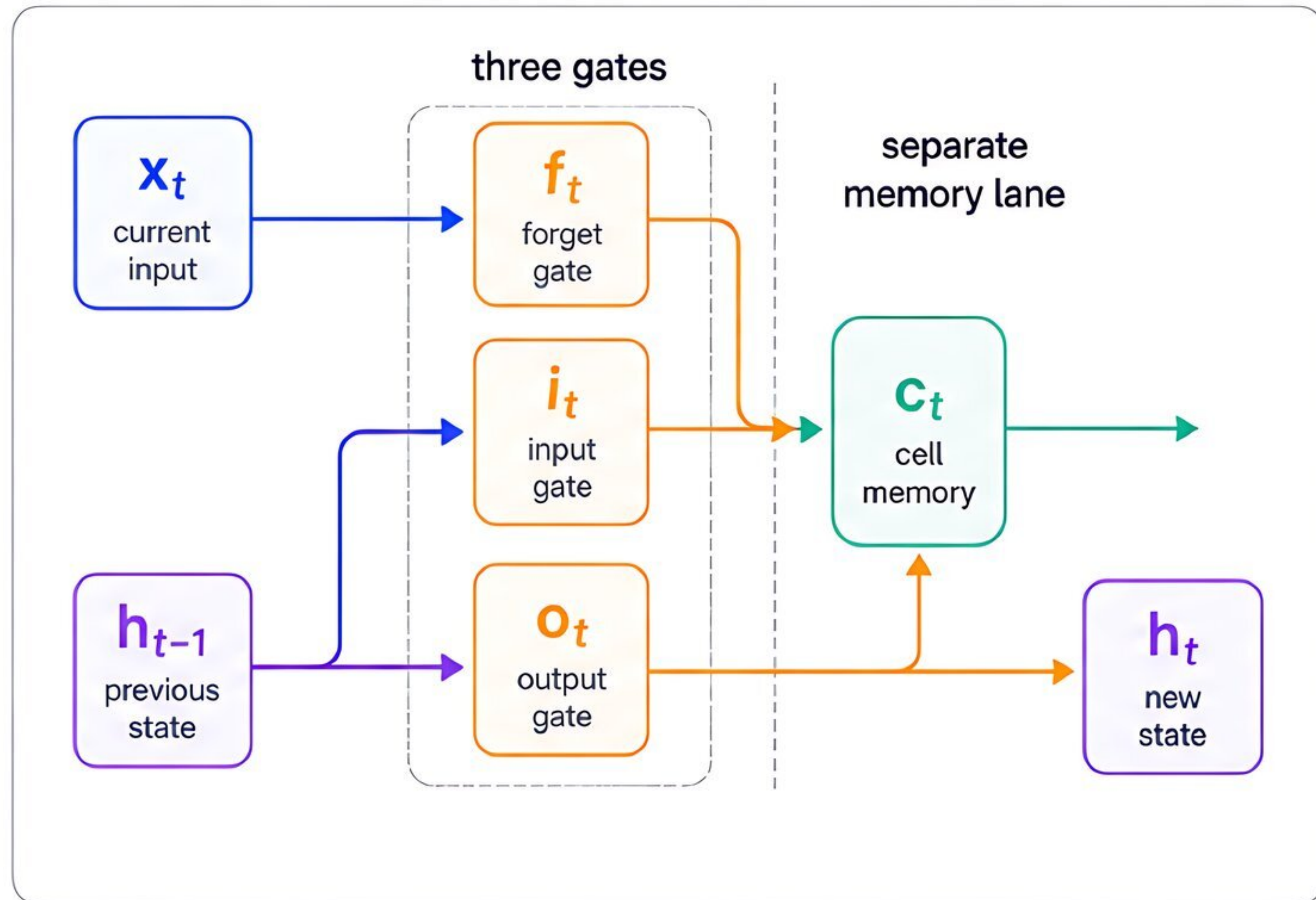
Gates protect useful information

Keep the battery clue. Drop the noise.

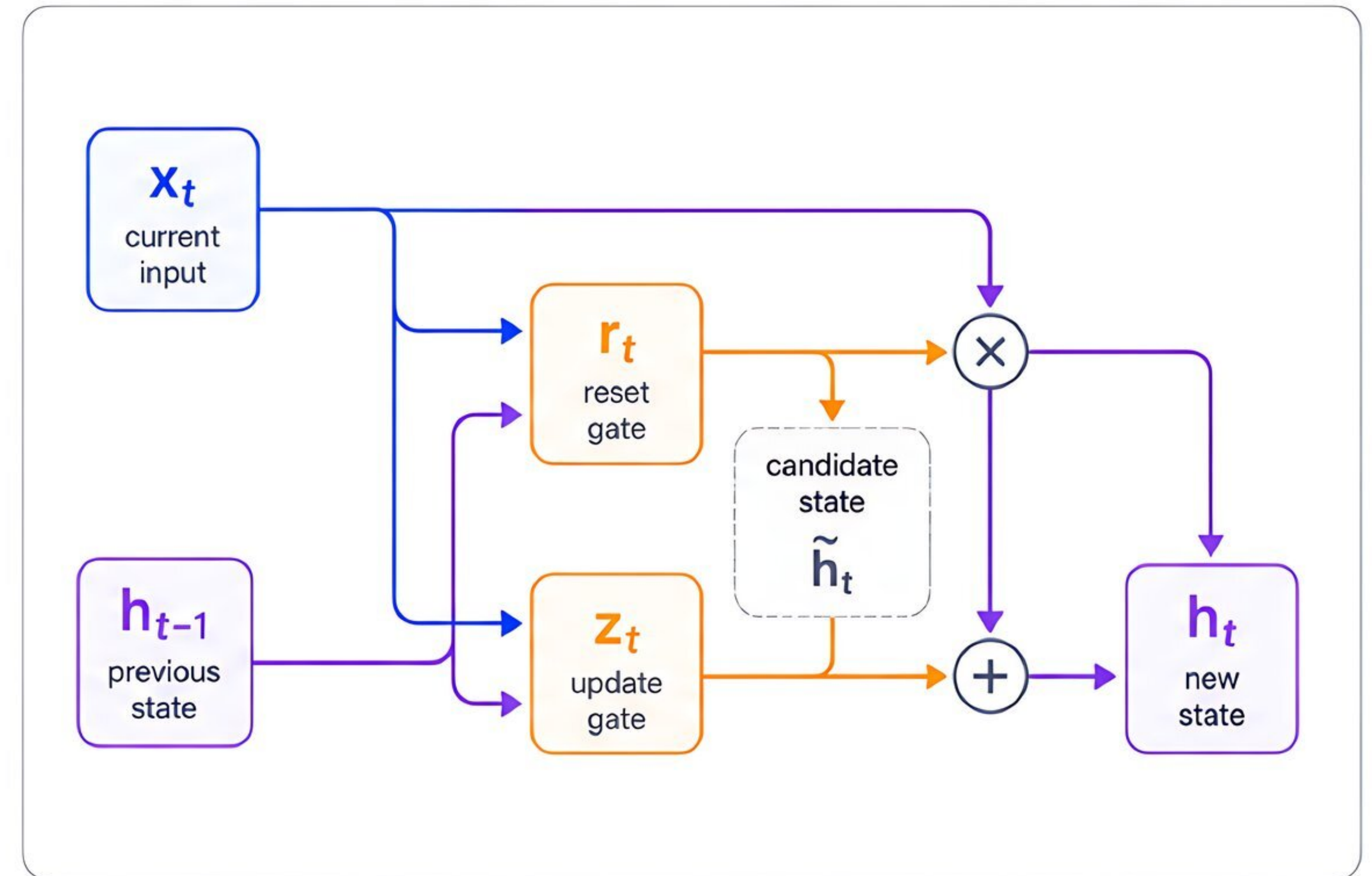


GRU: a leaner gated memory

Same idea, fewer controls



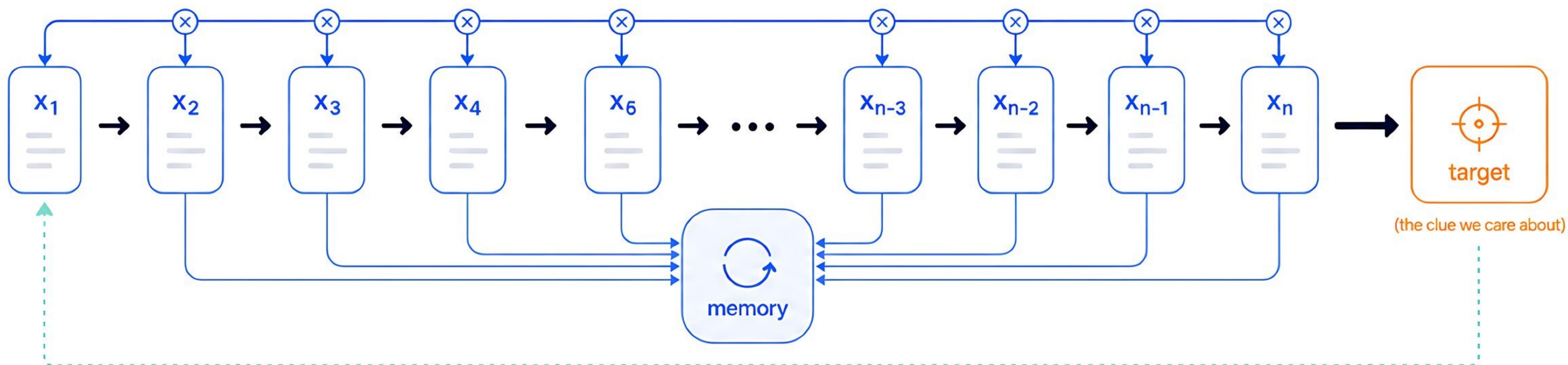
LSTM: more controls



GRU: fewer controls

Better memory is still not direct access

The clue still travels through every step.



What if we could look back?

⊗ gated memory path → sequential step → direct link (not available) 🎯 focus / target ○ output — token / context - - - limitation

A better question

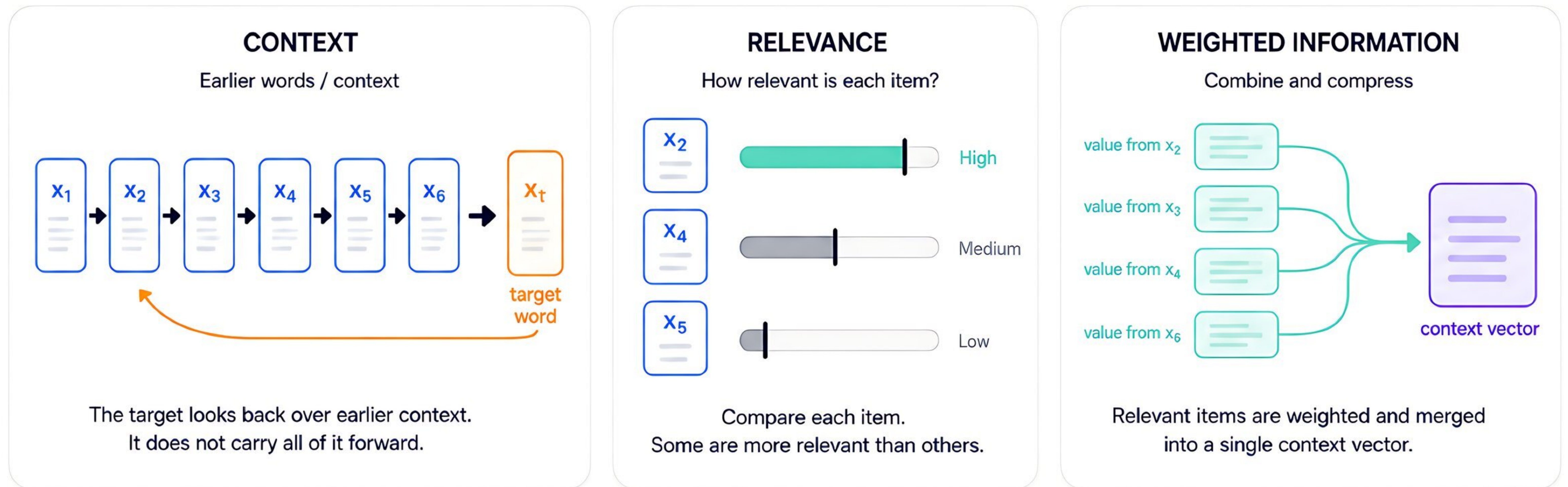
What information matters right now?



Choose relevant context.

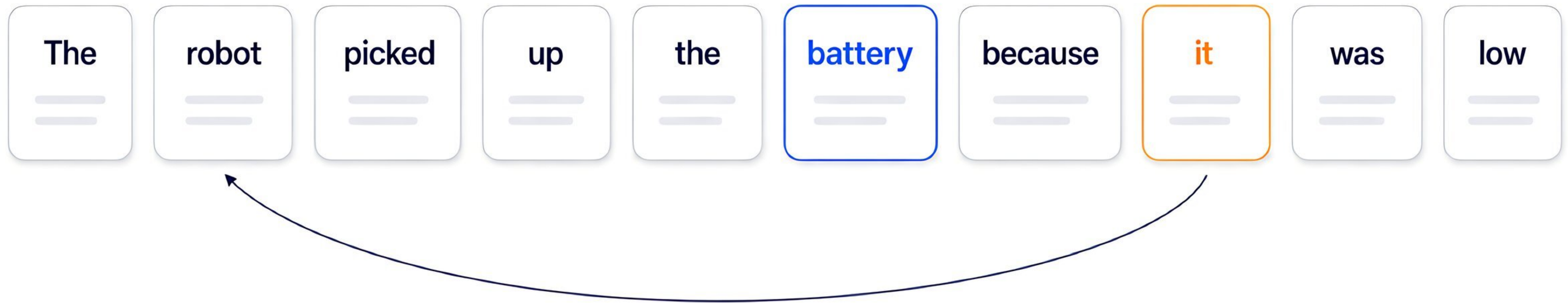
15 Attention is selective lookup.

Look back. Choose what matters. Bring it forward.



16 A running example.

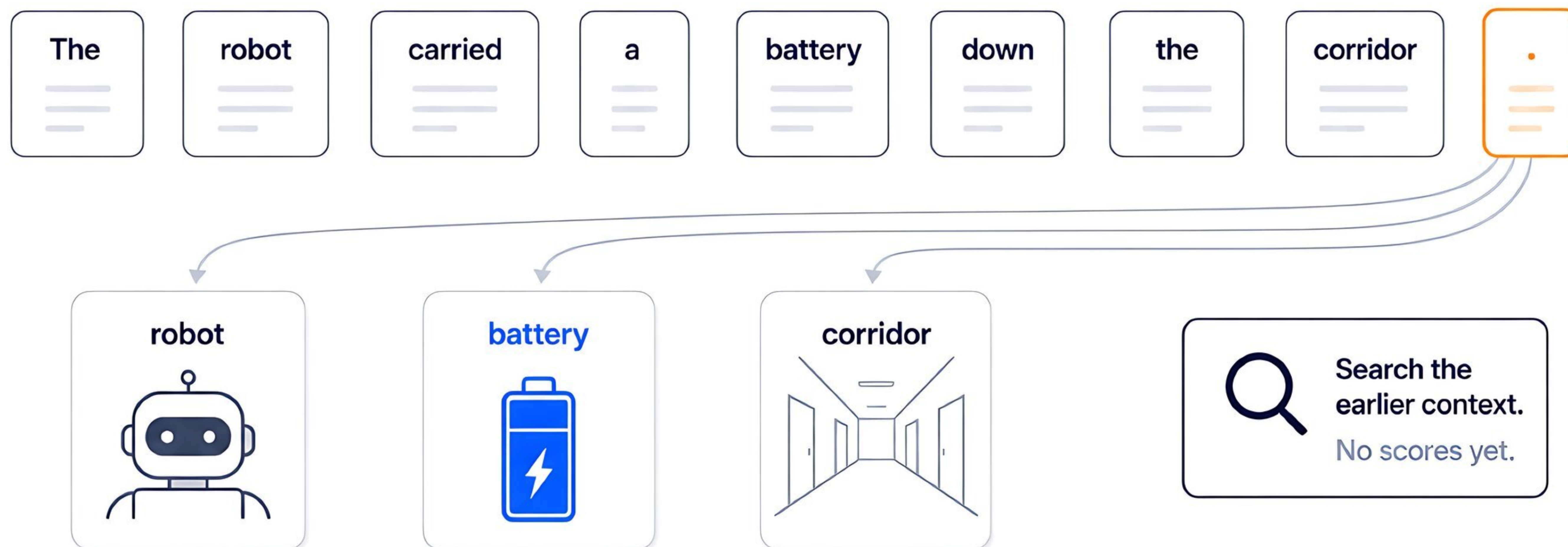
A small question with a big lesson.



What does “it” refer to?

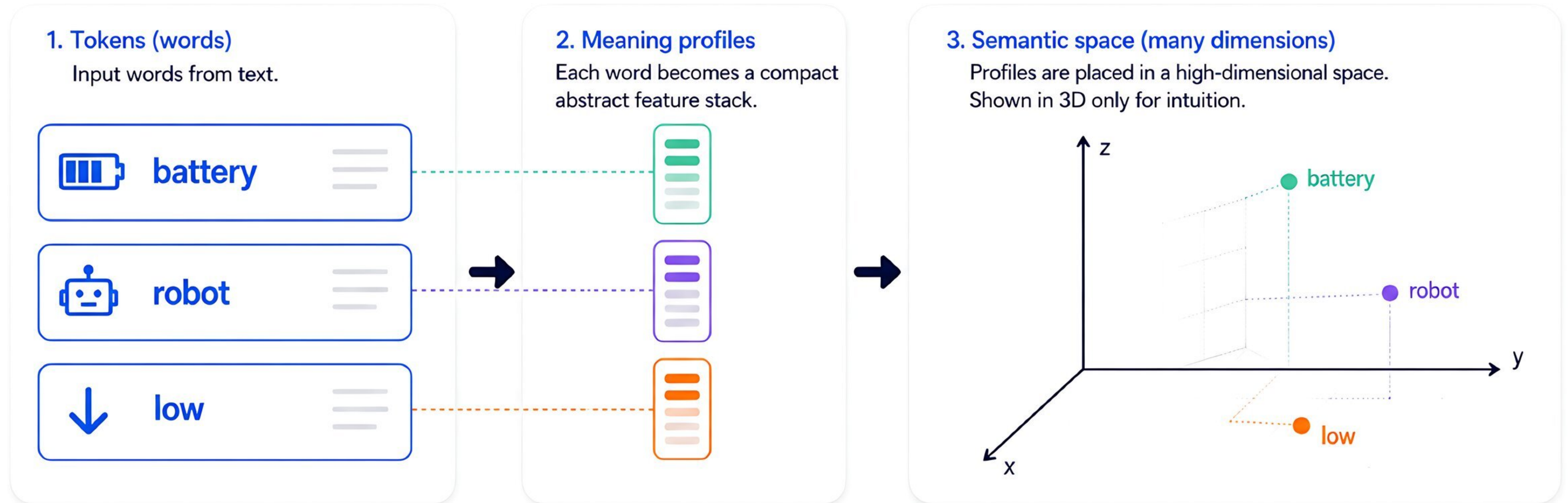
17 What should it look at?

Not every earlier word is equally useful.



18 Words become meaning profiles.

A learned list of numbers can capture useful patterns.



● Context (inputs) ● Relevant source ● Contextualized output ● Active focus ● Limitation / caveat ● Background

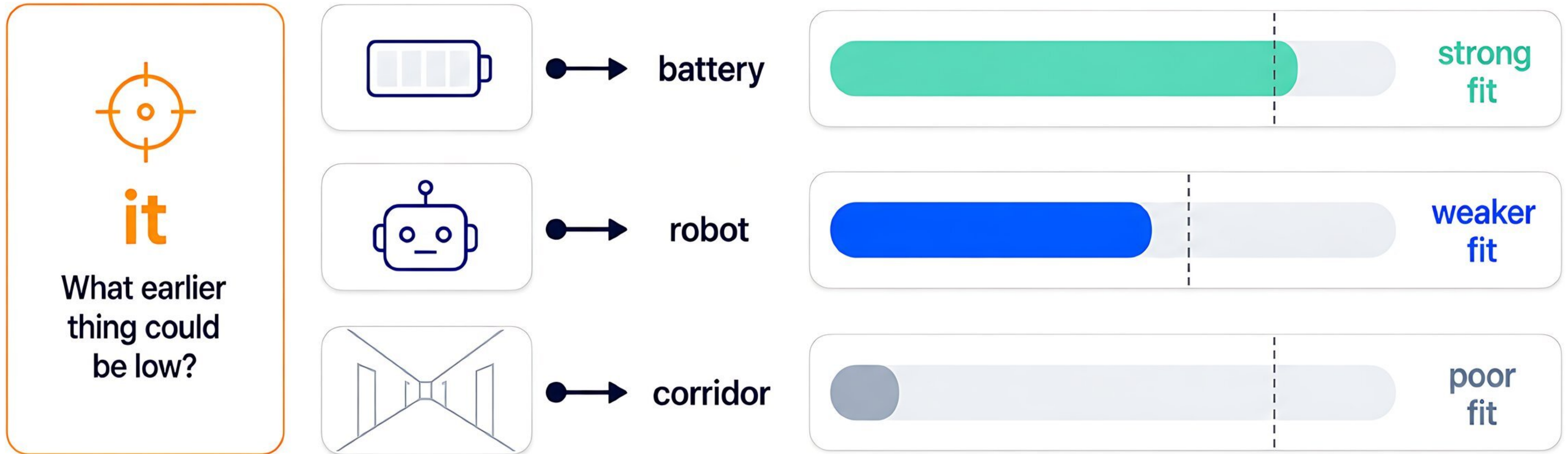


Key takeaway

Words are mapped to compact numeric profiles that capture patterns of meaning and relationships.

19 Compare what fits.

A stronger match earns a higher score.



Same direction = stronger fit.

20 Query.

What am I looking for right now?

 Query = the question this word asks.



Which earlier thing could be low?



21 Keys.

What kind of information does each word offer?



battery

object with charge



robot

agent



corridor

place



low

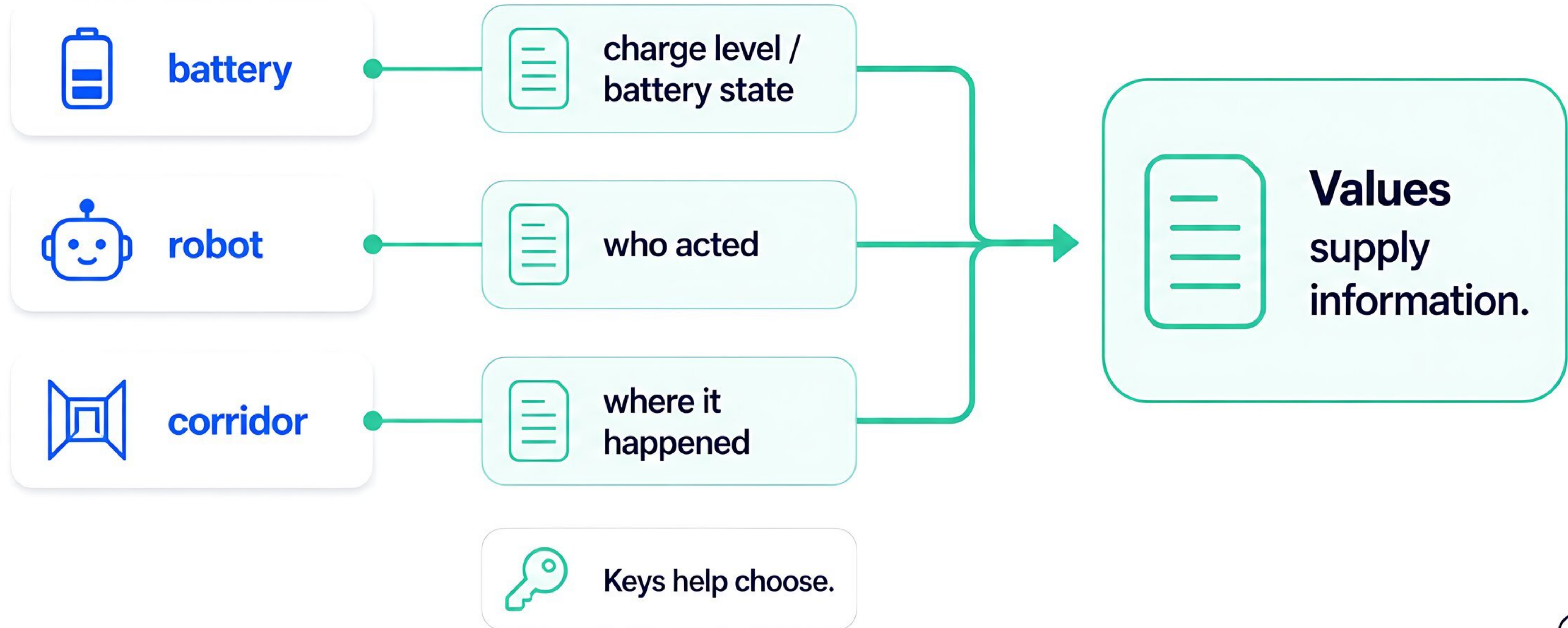
state



Keys are
searchable clues.

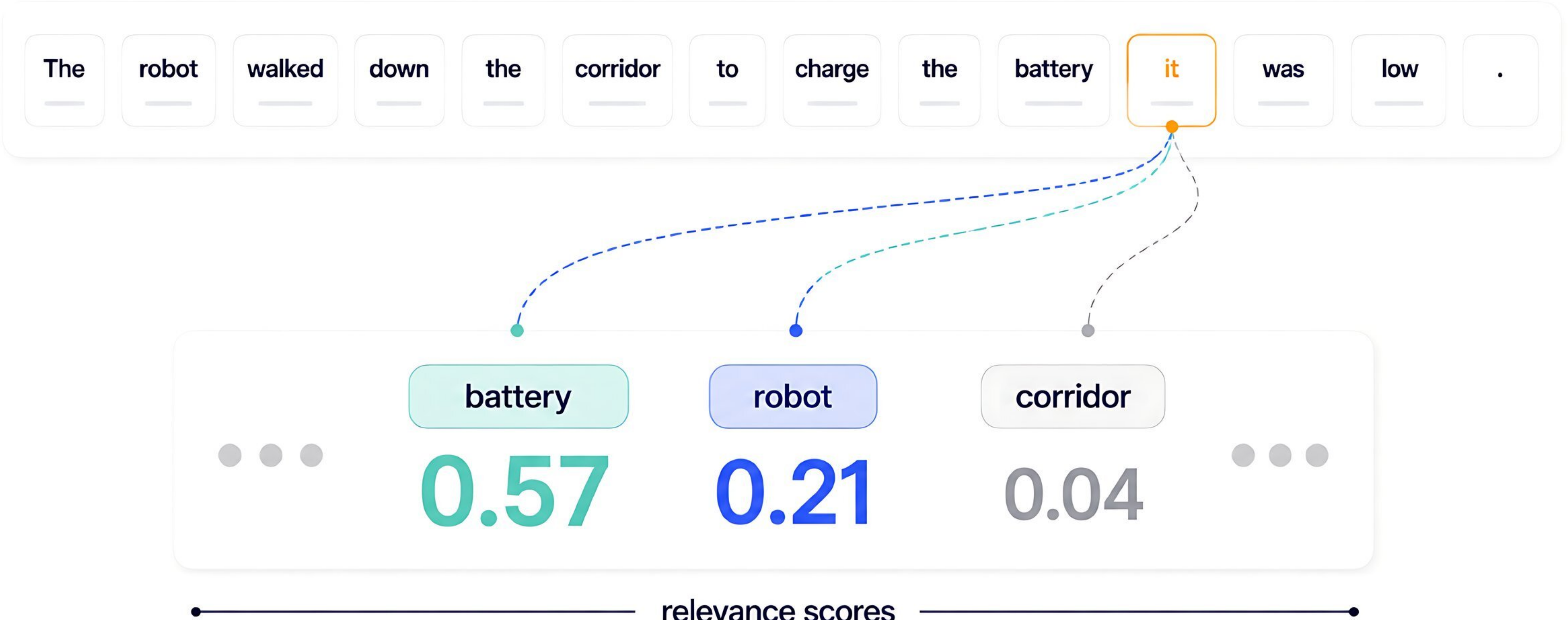
22 Values.

What useful information can each word contribute?



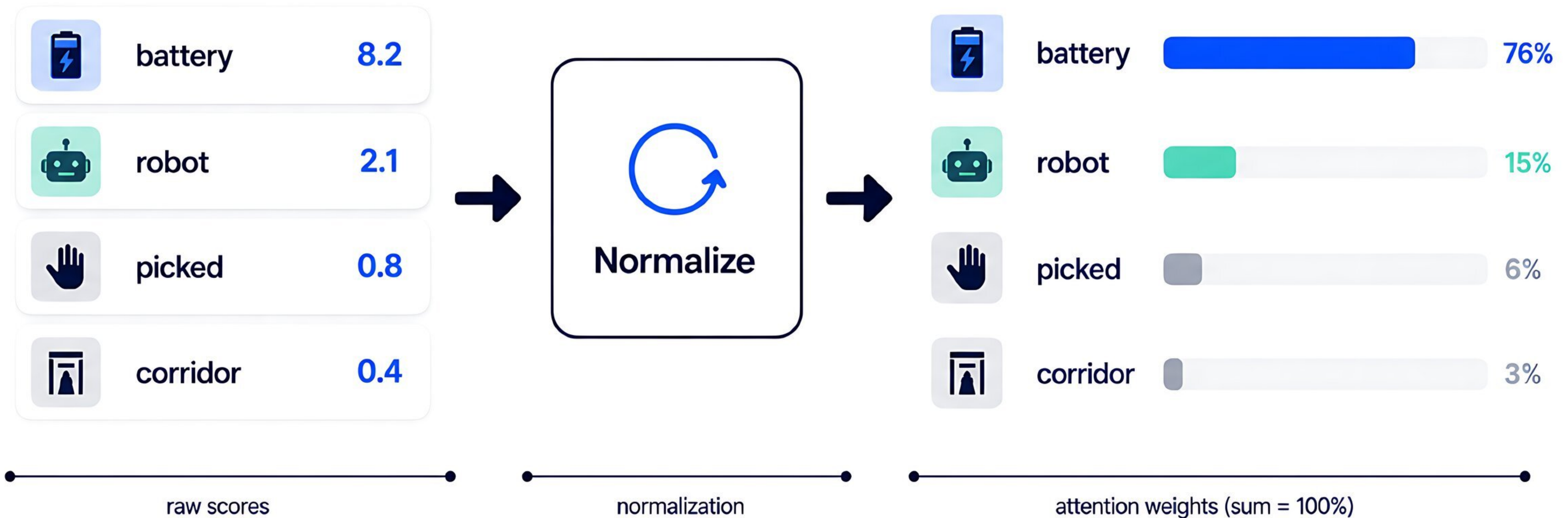
23 Attention score board.

Battery is the strongest match for it.



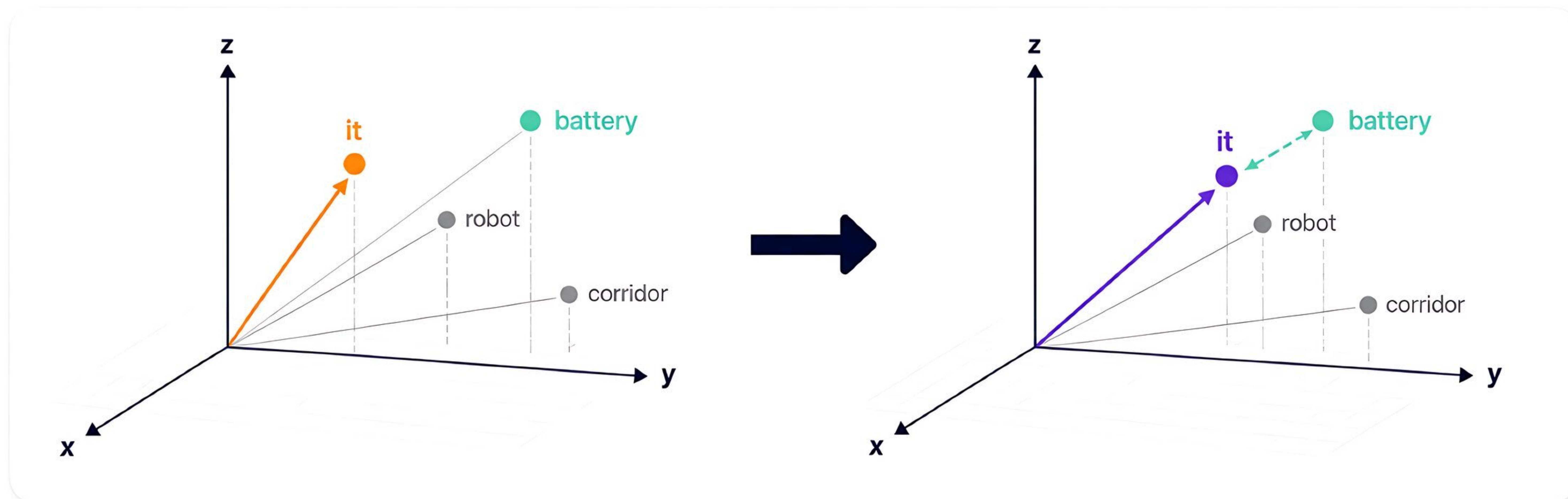
24 From scores to attention weights.

The strongest match gets most of the attention.



25 Build a context-aware meaning.

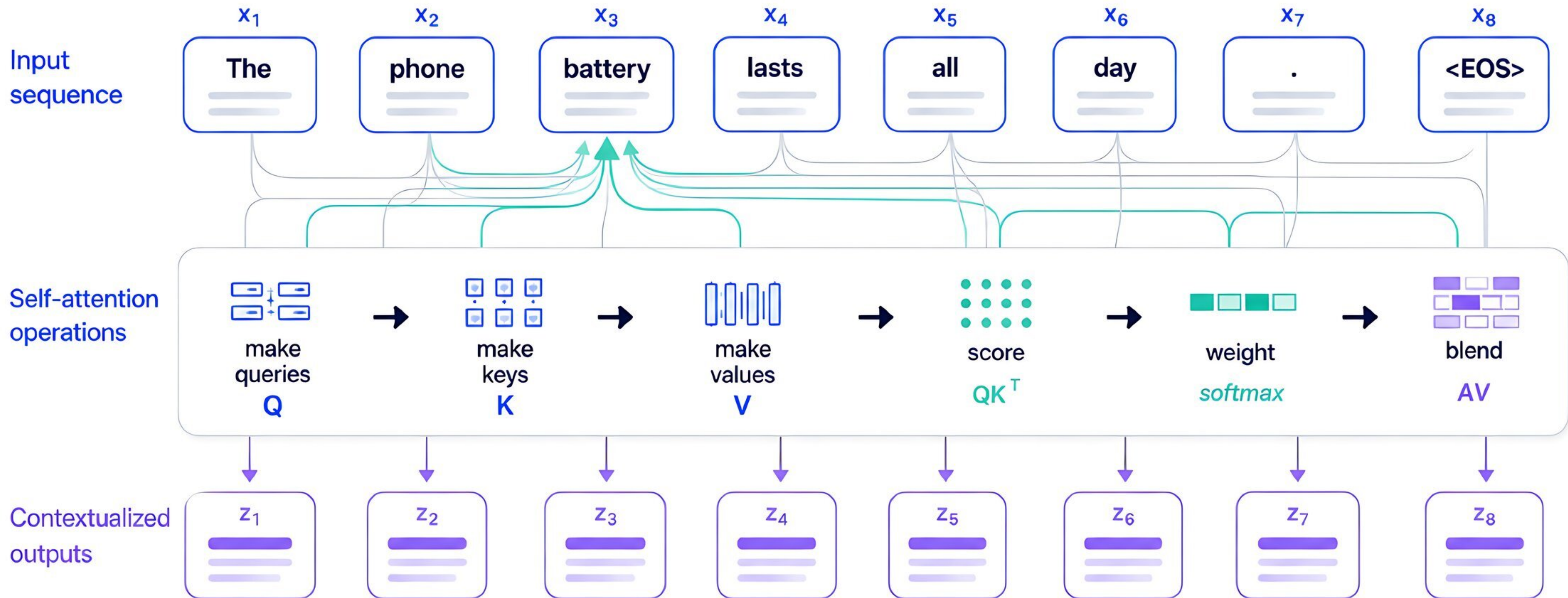
The word **it** now carries information about the **battery**.



The representation **changes** with context.

26 Self-attention.

Every word can look at the words that matter to it.



Next session: multiple views of attention and the Transformer.